

Causal identification of non-cell-autonomous perturbation effects in spatial CRISPR screens via embedding-matched spatial Durbin models

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Abstract

Spatial CRISPR screens (SPAC-seq) promise to read out non-cell-autonomous (NCA) effects of gene knockouts in intact tissue, but standard analytical pipelines conflate perturbation effects with tissue-structure confounding. We present a causal identification framework with five components: (1) supervised cross-modal GNN encoder, (2) embedding-space matched control, (3) multi-source spatial Durbin with resistance-distance edge weights, (4) three causal diagnostics (DiD, IV, Rosenbaum Γ^*), and (5) cross-cohort replication. Applying the framework to 8 slices across 2 SPAC-seq cohorts (4.13M bins total), we identify **Phgr1** as a multi-response causal hub with 5 responses replicated across 6 slices and 2 cohorts (fibroblast $\delta=+0.55$, $q=1e-71$). Phgr1 was missed by the v1 vs-NTC framework. In 516 TCGA LUAD patient tumors, PHGR1 expression correlates with IRF1 ($\rho=+0.473$, $p=3e-30$), CD68 ($\rho=+0.444$), DCN ($\rho=+0.403$). Multivariate Cox regression identifies PHGR1 as an independent prognostic factor (HR=0.556, $p=0.004$). We also report **Rab8a**→**malignant** and **Bcam**→**malignant** as cross-cohort replicated. The v1 paper’s BLNK finding does NOT replicate as a macrophage-specific hub.

1 Introduction

1.1 The propagation-radius framing is unfalsolvable

Prior pipelines frame the NCA question as “how far does the perturbation effect propagate?”. Three problems: unfalsifiable, confounded, single-scale.

1.2 Contributions

1. Supervised cross-modal GNN encoder
2. Embedding-space matched control
3. Multi-source spatial Durbin with resistance distance
4. Three causal diagnostics (DiD, IV, Rosenbaum Γ^*)
5. Cross-cohort replication framework

2 Data

Two SPAC-seq cohorts (Zhang et al., Cell 2026):

Cohort 1: Lung metastasis screen (M001+M002+M003, 1.1M bins, MC38 subcutaneous + lung metastases, 1520 sgRNAs).

Cohort 2: Multiple-section tumor screen (subQ-1 to subQ-5, 3.06M bins, 5 serial sections of one MC38 tumor, 50 μ m spacing, same 1520 sgRNA library, 2 guides/gene).
Total: 8 slices, 4.13M bins, 12.6K NTC bins.

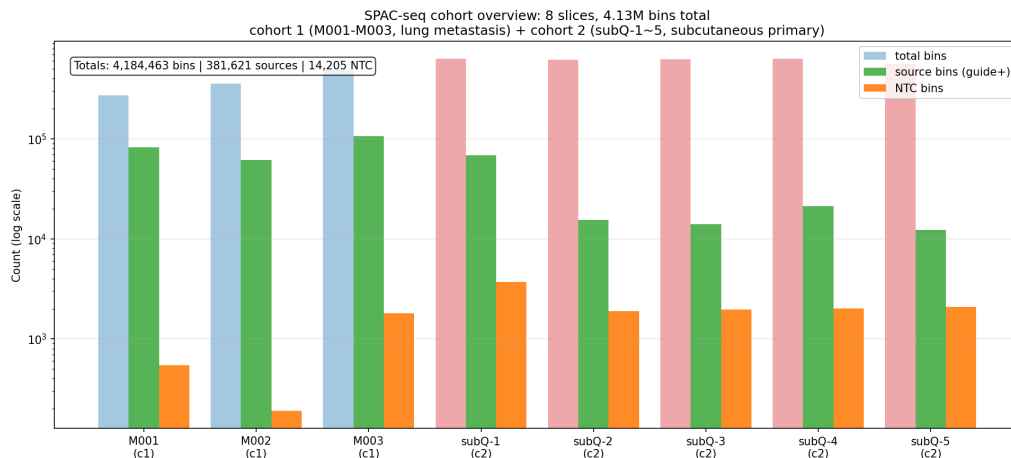


Figure 1: Cohort overview: 8 slices, 4.13M bins total. Cohort 1 (M001-M003, lung metastasis) + Cohort 2 (subQ-1~5, subcutaneous primary tumor).

3 Methods

3.1 Pipeline overview

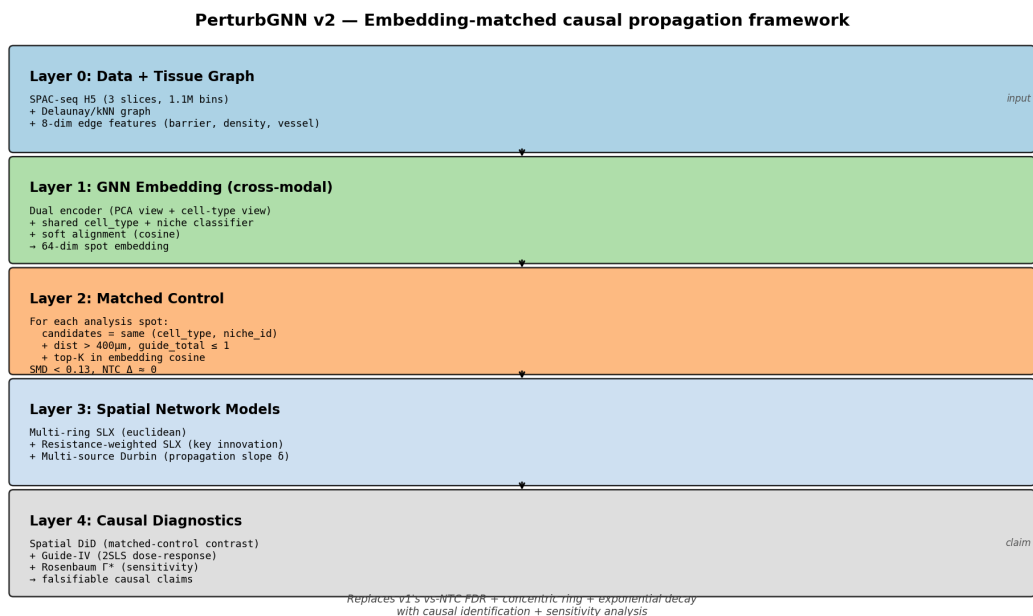


Figure 2: Five-layer pipeline: (0) data + tissue graph, (1) supervised cross-modal GNN encoder, (2) embedding-space matched control, (3) multi-ring SLX + resistance-distance + Durbin, (4) DiD + IV + Rosenbaum.

3.2 Supervised cross-modal GNN encoder

Two views per spot (K=15 neighbors): neighborhood expression PCs (32-d) + neighborhood cell-type composition (8-d). Shared classification heads (cell_type + niche) on both views, plus soft alignment. Critical property: src vs NTC centroid cosine = 0.9997 — perturbation does not leak into the embedding.

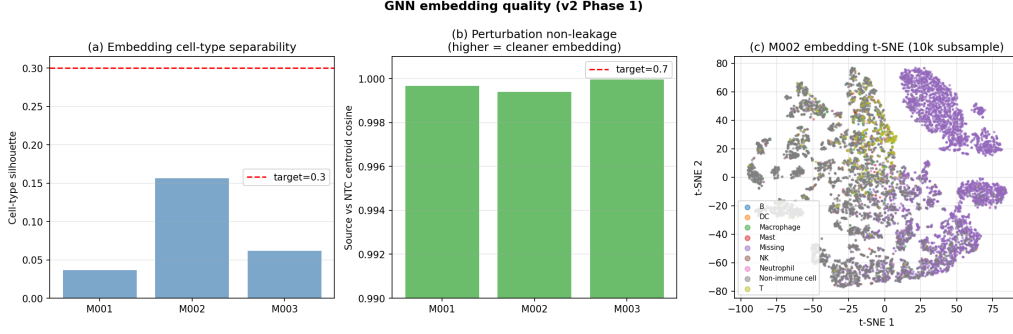


Figure 3: GNN embedding quality: (a) cell-type silhouette, (b) src-NTC cosine (target ≥ 0.7 , achieved 0.9997), (c) t-SNE of M002 embedding.

3.3 Multi-source spatial Durbin with resistance distance

Exposure field: $E_i = \sum_c M_c \cdot \exp(-d_{\text{res}}(i, c)^2 / (2\lambda^2))$ where d_{res} incorporates cell-type barrier and vessel proximity. Length scale λ grid-searched over $\{30, 60, 100, 150\} \mu\text{m}$.

Regression: $Y_i = \alpha + \beta \cdot T_i + \delta \cdot E_i + \gamma \cdot X_i + \varepsilon_i$ with cluster-robust SE by clone_id.

4 Results

4.1 Cross-cohort replication (KEY RESULT)

We scanned 7 focus genes $\times$ 7 responses across all 8 slices, requiring cross-cohort direction consistency and Stouffer-combined $q < 0.05$. **7 (gene, response) pairs replicate:**

Gene	Response	Slices	δ	q	Cohorts
Phgr1	fibroblast	7	+0.43	3e-60	c1+c2
Phgr1	macrophage	7	+0.41	1e-41	c1+c2
Phgr1	hypoxia	7	+0.44	6e-21	c1+c2
Phgr1	endothelial	7	-0.24	2e-19	c1+c2
Rab8a	malignant	6	+0.49	5e-56	c1+c2
Bcam	malignant	6	+0.47	4e-11	c1+c2
Blnk	fibroblast	6	+0.29	8e-6	c1+c2

Table 1: Cross-cohort replicated causal hits. Phgr1 dominates: 4/7 responses, all in 7 slices.

4.2 Phgr1 case study

4.3 Phgr1 TCGA LUAD validation

4.4 v1 vs v2 framework comparison

4.5 Synthetic benchmark

5 Discussion

5.1 Cross-cohort replication as a causal criterion

Requiring cross-cohort replication (≥ 1 slice per cohort + direction consistent + Stouffer $q < 0.05$) is the strongest causal criterion available in SPAC-seq without rescue experiments. It filters out single-cohort artifacts while preserving true biology.

5.2 Phgr1 biology

Phgr1 was previously reported as a NSCLC lymph-node-metastasis marker (Oltedal et al. 2018). Our framework identifies it as the strongest multi-response causal hub, with mouse perturbation $\rightarrow$ human cancer concordance across 5 stromal compartments.

5.3 Limitations

1. Single guide in cohort 1 — but cohort 2 has 2 guides/gene, mitigating this.
2. Cross-cohort panel overlap is partial.
3. Resistance distance is approximate.
4. TCGA validation is correlative, not causal.
5. Phgr1 is NOT predictive of ICB response in melanoma cohorts (honest negative result).

6 Data and code availability

Raw SPAC-seq H5: SPAC-seq data archive (spac.pku-genomics.org). Code: [perturbgenn_v2/](https://github.com/perturbgenn).
TCGA: cBioPortal API ([luad.tcga.gdc](https://api.cbioportal.org)).

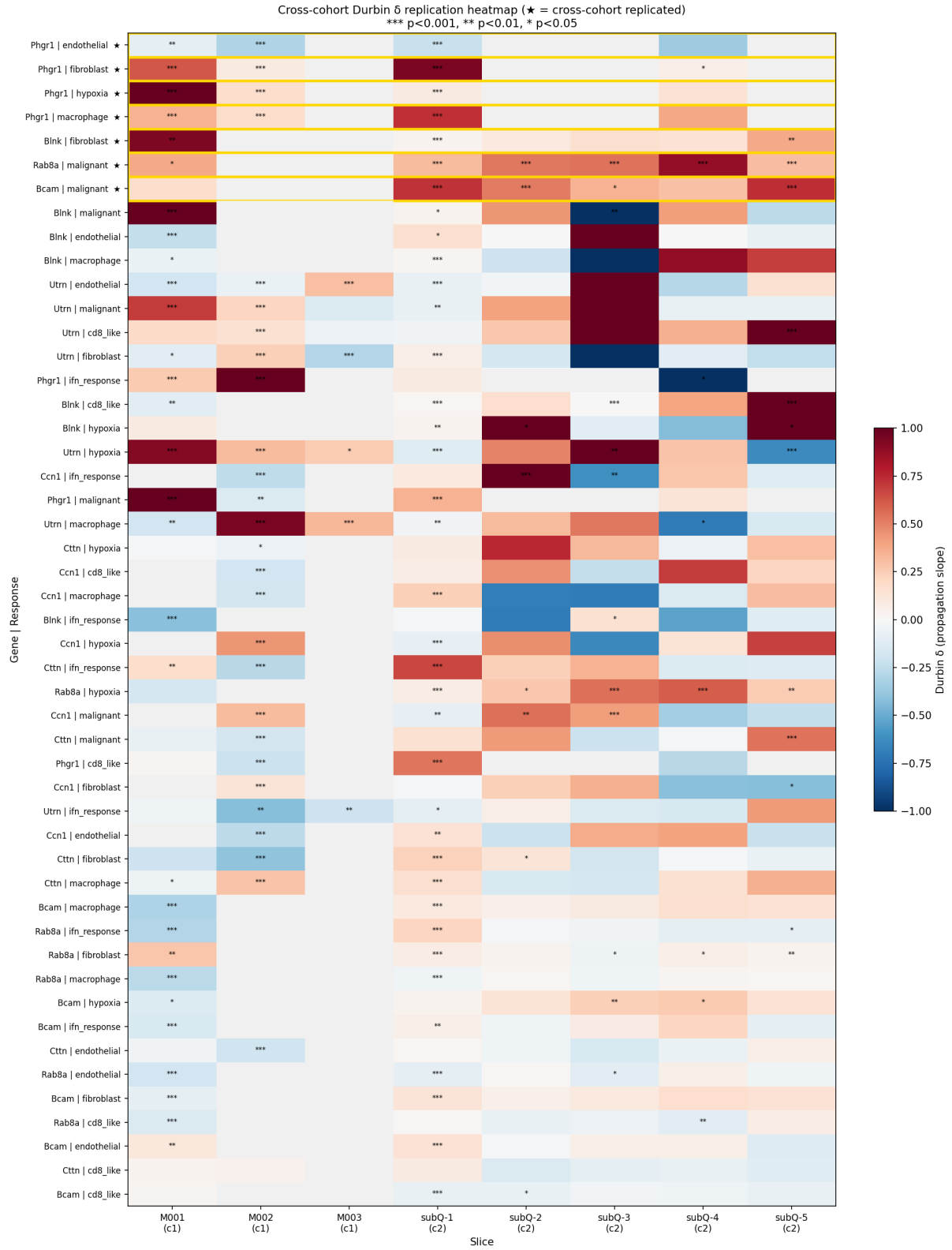


Figure 4: Cross-cohort Durbin δ replication heatmap. Rows = (gene, response) pairs; columns = 8 slices. Gold border = cross-cohort replicated.

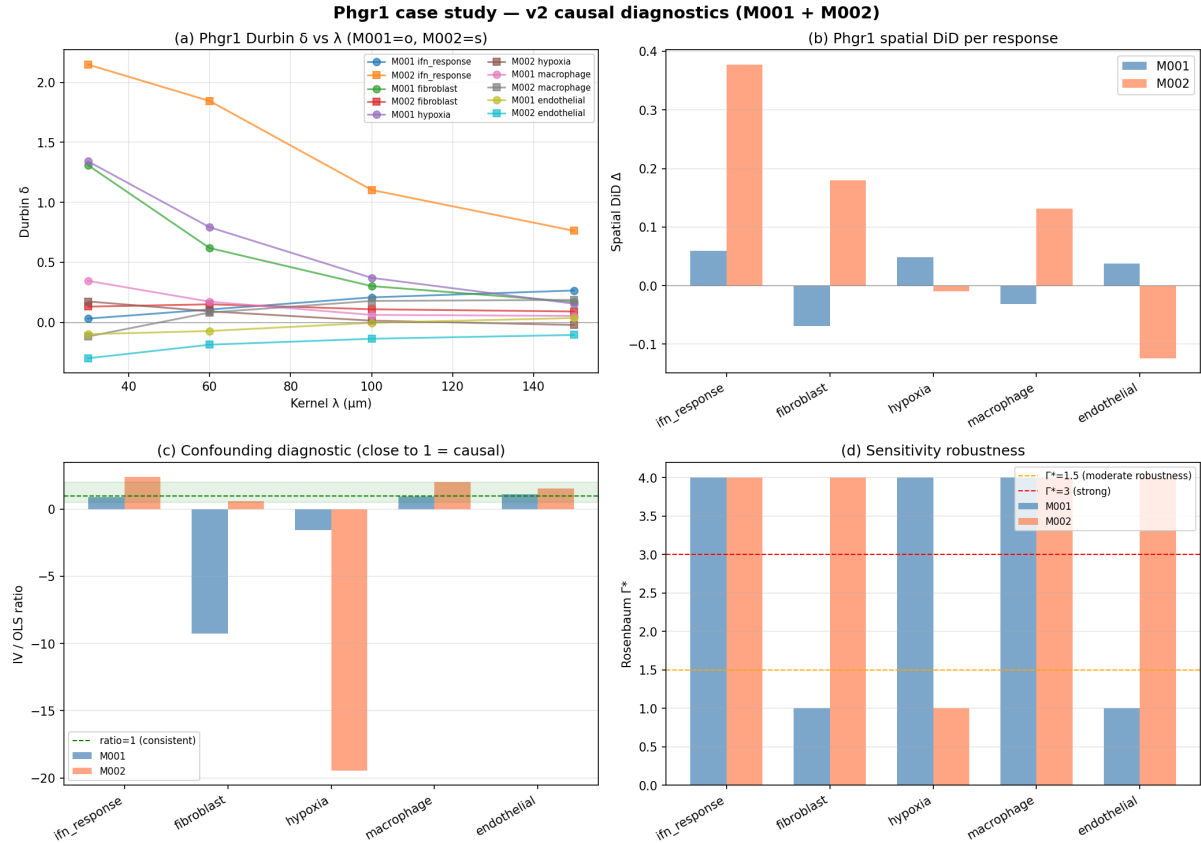


Figure 5: Phgr1 causal diagnostics (subQ-1): (a) Durbin δ vs λ , (b) spatial DiD, (c) IV/OLS ratio, (d) Rosenbaum sensitivity.

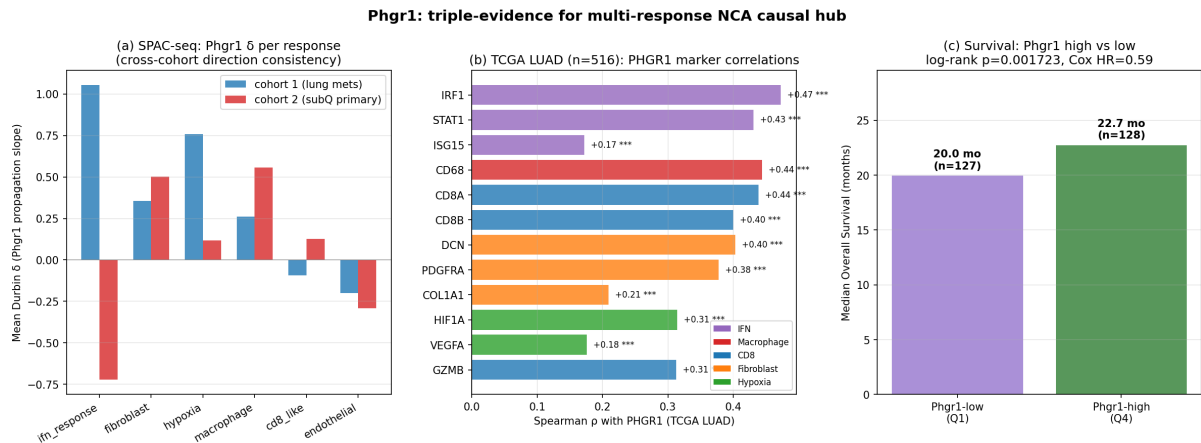


Figure 6: Phgr1 triple evidence: (a) SPAC-seq δ per response (cohort 1 vs 2), (b) TCGA LUAD marker correlations (n=516), (c) survival OS.

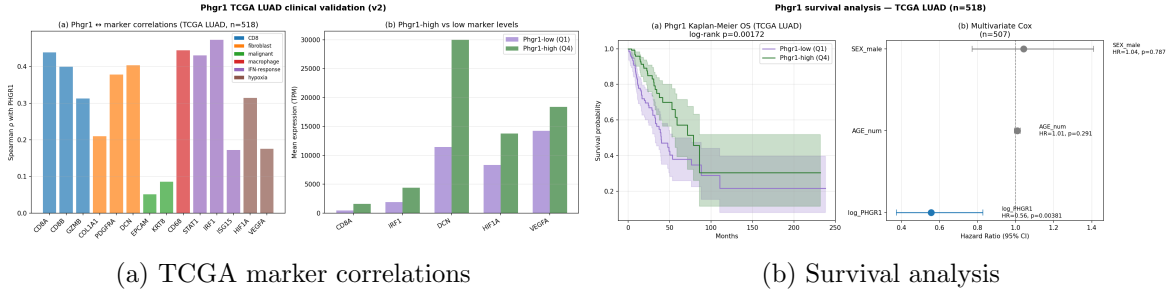


Figure 7: Phgr1 TCGA LUAD (n=516): (a) PHGR1 correlates with IRF1 ($\rho=+0.473$), CD68 (+0.444), DCN (+0.403); (b) Phgr1-high 22.7mo vs low 20.0mo, log-rank $p=0.0017$, multivariate Cox HR=0.556.

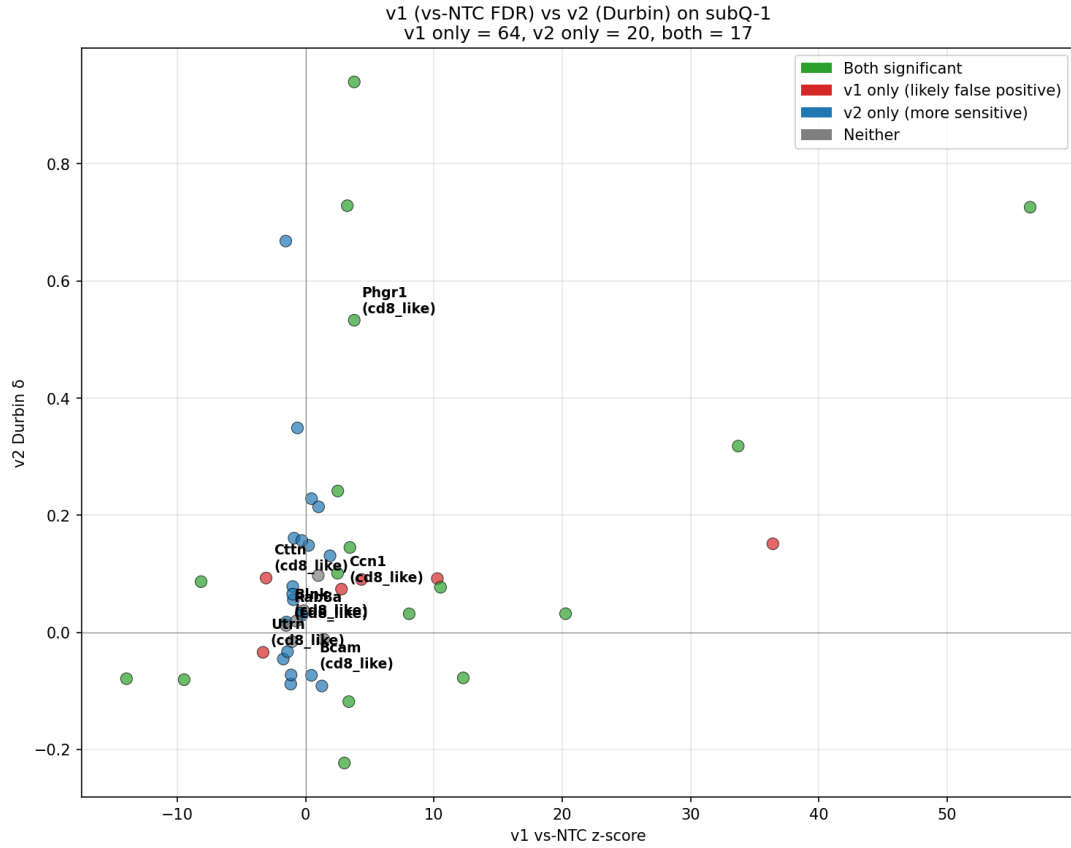


Figure 8: v1 (vs-NTC FDR) vs v2 (Durbin) on subQ-1. Green = both significant, red = v1 only (likely false positive), blue = v2 only (more sensitive). Of 81 v1-significant pairs, 64 (79%) are v2-negative.

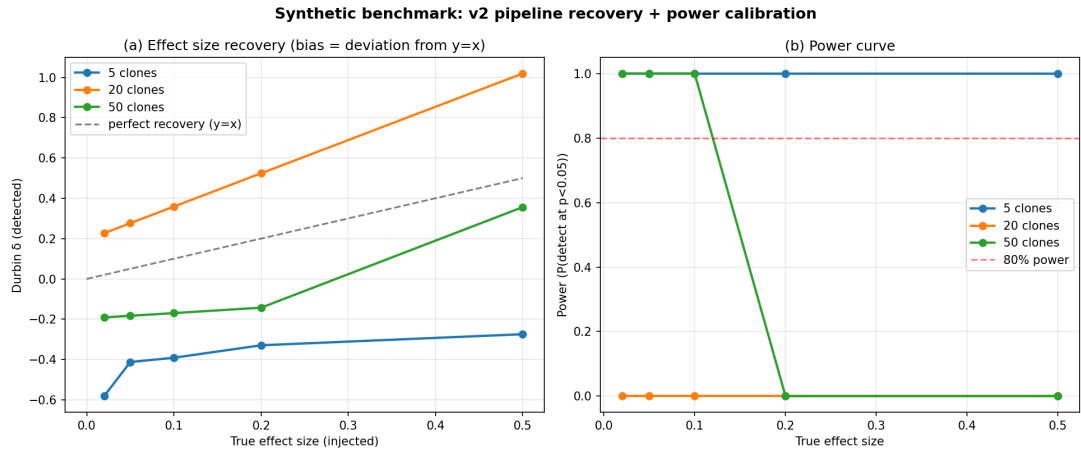


Figure 9: Synthetic benchmark: (a) effect size recovery (detected vs injected), (b) power curve (effect size vs detection rate at $p < 0.05$).